Tower-Coordinate-Free Road-Level Trajectory Recovery from Cell-ID Sequences: Supplementary Material

Xiaoxiao Wu*, Xiao Zheng*, Hairong Zhou*, Yongjie Zhu†
* College of Electronics and Information Engineering, Shenzhen University, Shenzhen, China
Emails: xxwu.eesissi@szu.edu.cn, m15649741159@163.com, zhr19466@gmail.com
† Nexwise Intelligence China Limited, Guangzhou, China
Email: zhuyj@nexwise.com.cn

Reviewer Navigation

This concise supplement contains the material most useful for checking implementation and empirical claims: **A**, the information boundary, evidence artifact, graph contract, and metrics; **B–E**, the exact PPO state, reward, optimization, and training-only oracle built around Set-Valued Asynchronous Alignment (SVAA); **F**, online time-aware SVAA/PPO-guided MCTS; **G**, controlled baseline and ablation definitions plus aggregate results; **H**, method and qualitative figures; and **I**, the separate Region B evaluation together with data-efficiency and repeatability audits. **J** gives the reviewer checklist and limitations. The full 13-sheet route gallery and per-trajectory tables are excluded from this reviewer-facing PDF; the all-trajectory ablation audit is retained as a fixed-order heatmap.

Terminology. We call the proposed sequence module *Set-Valued Asynchronous Alignment (SVAA)*. SVAA computes a maximum-cardinality ordered matching from Cell-ID observation positions to nonempty, nonoverlapping, contiguous path-edge intervals. A set-valued LCS recurrence is only its deterministic anchor step; SVAA then expands compatible intervals without overlap. Unlike ordinary LCS, compatibility is many-to-many rather than token equality; unlike synchronous matching, SVAA permits skipped observations/edges and lets one observation explain a multi-edge interval. Throughout this supplement, *SVAA witness* means this deterministic output. The main paper's "CID/LCS progress" features are precisely its cardinality and latest increment.

A. Inputs, Evidence, and Deterministic Graph Contract

Information boundary.: Regional calibration uses the probe's GPS–CID records, session/trace identifiers, original acquisition order, and a frozen directed road graph. The supplied coordinates have already been de-duplicated. Evidence-cache construction, training-data generation, the training-only coordinate oracle, PPO training, and bounded training search use no timestamps. Online prediction receives the frozen graph, endpoints, a timestamped raw-CID sequence, evidence cache, PPO checkpoint, and matching normalizer. Query timestamps determine the online hard temporal constraints and soft pair-time/corridor terms used by action priors, leaf evaluation, and final ranking. Intermediate query GPS and reference edges are withheld from prediction. Held-out test records are reserved for final evaluation.

Raw-CID token and sequence construction.: The model token is the normalized raw CID alone. Public land mobile network (PLMN) and radio access technology (RAT) are retained only for acquisition audit, not as model inputs or cache keys. In Region A, 295 of 911 raw-CID values occur under multiple PLMN namespaces, all using Long-Term Evolution (LTE); the system intentionally pools records sharing a raw CID into one empirical support Ω_c. It neither asserts one physical cell nor performs operator identification or cross-operator cell disambiguation. Each retained record is the `Index=1` cell from one logical scan channel—the channel's first-ranked scan result, not an asserted serving or camped cell. Records form one sequence. Equal-time records retain their deterministic device-output order and equal timestamp, without implying physical switch chronology. All CID-consuming methods receive the same permitted raw query information. Yen, Beam, and the proposed decoder share the query-sequence and evidence contract above; other baselines may apply their documented internal pooling or subsampling. Common raw information therefore does not require identical internal representations. No rank weighting or multi-stream fusion is used by the proposed decoder.

A.1 Road–Cell-ID Evidence Artifact

Accepted calibration rows contain CID, longitude, latitude, and either a trace identifier or a single-trace-per-file declaration. Coordinates have already been de-duplicated in the supplied base data; repeated CID labels at different retained positions remain distinct records. Malformed or nonfinite required fields are errors. Processing uses fixed trace and original within-trace row order for reproducible reductions and geometric displacements. No timestamp is required or used for sorting, grouping, coordinate de-duplication, filtering, or evidence aggregation.

Let (λ_0, ϕ_0) be the mean accepted longitude and latitude in degrees. Map every point and road vertex to metres by

$$x = (\lambda - \lambda_0)\cos(\pi\phi_0/180)\,111320,$$
$$y = (\phi - \phi_0)\,111320.$$

Thus the trigonometric argument is in radians while longitude and latitude differences remain in degrees. For CID c, bin points into 80-metre cells, retain cells in decreasing count (then increasing cell index) through at least 90% cumulative mass, and buffer their union by 50 metres to form the closed support Ω_c. With population variances over all N_c records,

$$\sigma_c = \min\{800, \max\{80, \sqrt{\mathrm{Var}(x) + \mathrm{Var}(y)}\}\}\ \mathrm{m}.$$

Only $N_c \geq 2$ CIDs are scored. Region A has 92,232 raw records and 911 normalized raw-CID values; ten singleton values are excluded, leaving 92,222 records supporting 901 scored raw-CID values.

Direction uses geometric displacement in the original acquisition order of the already coordinate-de-duplicated records. For consecutive retained projected sampling positions $\boldsymbol{p}_j, \boldsymbol{p}_{j+1}$ within the same trace, retain a displacement of at least 1 m and form

$$\boldsymbol{u}_j = \frac{\boldsymbol{p}_{j+1} - \boldsymbol{p}_j}{\|\boldsymbol{p}_{j+1} - \boldsymbol{p}_j\|_2}.$$

Assign the vector to a CID observed at both positions. No vector crosses a trace boundary. No timestamp, elapsed-time threshold, or speed filter is used. For CID c, let $\mathcal{U}_c$ be its assigned unit displacements. When $\mathcal{U}_c$ is nonempty, define $\bar{\boldsymbol{u}}_c = |\mathcal{U}_c|^{-1} \sum_{\boldsymbol{u} \in \mathcal{U}_c} \boldsymbol{u}$, and $R_c = \|\bar{\boldsymbol{u}}_c\|_2$. Direction is reliable exactly when $|\mathcal{U}_c| \geq 3$ and $R_c \geq 0.50$; in that case $\mathrm{dir}(c) = \bar{\boldsymbol{u}}_c / R_c$. Otherwise the directional component is omitted as specified in Table S1.

Canonical source-directed edges are sampled at 5-metre arc-length intervals, including both endpoints. With sample set $Q(e)$,

$$S_d(e, c) = \exp[-d(e, \Omega_c)^2/(2\sigma_c^2)],$$
$$S_h(e, c) = |\{q \in Q(e) : \Omega_c \text{ covers } q\}|/|Q(e)|,$$
$$S_o(e, c) = \{1 + \mathrm{dir}(e)^{\mathsf{T}}\,\mathrm{dir}(c)\}/2.$$

Table S1 completes the operational contract. An unlisted edge–CID pair has score zero.

Graph and endpoint embedding.: For each ordered node pair, retain the shortest directed edge and break an exact tie by the lexicographically smallest original edge identifier. Order outgoing edges by original edge identifier, target, and canonical identifier. Actions occupy 13 fixed slots; absent and hard-infeasible slots are masked, and graphs with larger canonical out-degree are rejected. Start is embedded before goal by deterministic projection onto the successively updated directed graph. Projection ties choose the earliest geometry segment and then the smallest canonical edge identifier. A reverse edge is also split only when its independent projection lies within 5 m. Virtual connector segments add no off-road length and inherit their own source-directed edge identity.

A.2 Evaluation Protocol and Edge Cases

Evaluation removes virtual connectors and restores canonical directed source edges. Road-edge precision/recall use shared predicted/reference edge identities; length-weighted variants replace counts with the corresponding edge lengths. F1 is the harmonic mean. A failed recovery contributes zero to all overlap metrics. $S_{\mathrm{OD}} = 1$ only for a nonempty, directed-contiguous path connecting the embedded endpoints. All metrics are computed per trajectory from unrounded identities and lengths and then macro-averaged over all 250 held-out queries.

Paired confidence intervals resample the same 250 query indices with replacement for both methods 10,000 times and take the 2.5 and 97.5 percentiles of the macro-score difference.

B. Exact 20-Component SVAA-Aware PPO State

After every appended edge, recompute the SVAA witness. The next pointer is the first observation after the final match, or a completion sentinel. The local observation window spans two positions before through eight after that pointer. Let the distance scale be the larger of 1 m and initial straight-line start–goal distance. Table S2 defines the raw state; every component is clipped to $[-1, 1]$, normalized with the saved training mean and variance plus 10^{-8}, and clipped to $[-10, 10]$.

C. SVAA-Aware Training Reward and Episode Boundaries

For a valid step, sum Table S3 and clip to $[-8, 12]$. Lookahead and bounded-search terms have total-reward weights 0.8 and 0.5; other terms have unit weight. An invalid, masked, or dead-end action instead returns exactly -1 and terminates.

D. PPO Optimization and Artifact Binding

Stop the counter exactly at 5,000,000 interactions. Count a final incomplete collection but do not update from it. Region A and B share one fixed evidence, graph/action, state, reward, PPO, search, temporal, corridor, and ranking configuration. They differ only in regional records, graphs, generated-training and held-out-test sets, fingerprints, and learned PPO/normalization values. For each region, model fitting uses only its 1,000 generated training trajectories. Development-time hyperparameter selection uses only the Region A generated corpus; none of the 250 real test trajectories is used for model selection. Region B reuses this Region-A-selected configuration without region-specific retuning.

E. Training Oracle and Bounded Lookahead

Calibration-derived policy-training trajectories.: All reported training trajectories are generated offline from the frozen regional calibration corpus and road graph, without additional training drives. For each region, we randomly select 1,000 O–D coordinate pairs from the calibration data, generate a route and longitude–latitude sequence on the graph, and match every generated coordinate against the calibration records to assign its normalized raw-CID token. The resulting GPS^*–CID sequences supervise PPO; GPS^* denotes generated route coordinates, not measured GPS from a training drive. Sufficient

TABLE S1
OPERATIONAL ROAD–CELL-ID EVIDENCE AND ARTIFACT CONTRACT.

Item	Deterministic rule
Normal evidence	$E_{\mathrm{CID}} = (1.0S_d + 0.8S_h + 1.8S_o)/3.6$. For an unreliable CID direction, set its direction weight to zero and use $(1.0S_d + 0.8S_h)/1.8$.
Compatibility	(e, c) is compatible iff cached $E_{\mathrm{CID}}(e, c) > 0.18$; the strict comparison uses the unrounded value.
Sparsification	For every $N_c \geq 2$ CID, rank all frozen source edges by decreasing unrounded score and then canonical edge identifier; keep $\min(300, \lvert E\rvert)$.
Geometry	Use the frozen, source-oriented geometry and exact 5-metre samples. Boundary samples count as support hits. Reject zero-length projected geometry.
Artifact binding	Require schemas `paper-ecid-v1`/`paper-ecid-raw-v1`, generator constants, projection origin, per-CID parameters, sorted score map, normalized-record hash, graph and source-edge hashes, generator hash, and software versions. Query split edges inherit only their source edge's score map.
Validation	Reject missing or nonfinite metadata, noncanonical/conflicting edge keys, more than 300 source edges for a CID, graph/fingerprint mismatch, or any score outside $[0, 1]$.

TABLE S2
EXACT RAW 20-COMPONENT PPO STATE BEFORE CLIPPING AND NORMALIZATION.

No.	Feature	Construction
1	Goal distance	Divide by the distance scale, cap at one, map to $[-1, 1]$.
2–3	Goal bearing	Normalized longitude and latitude components of the current-to-goal vector.
4	Direction similarity	Cosine between latest movement and goal direction; zero initially.
5–6	SVAA progress	Witness cardinality divided by observation count, and latest cardinality increment divided by that count.
7	Search depth	Current edge count divided by observation count.
8	Last-edge match	Fraction of the latest edge's compatible labels also present as distinct values in the local window; zero initially.
9	Out-degree	Canonical out-degree divided by the frozen graph's actual maximum canonical out-degree.
10–12	Path status	Step/1500; path length/distance scale; and repeated nodes among the latest 12 nodes divided by 12.
13	One-step potential	Maximum distinct observed compatible-label count over canonical successors, multiplied by 0.2 and capped at one.
14	Three-step coverage	Breadth-first search to depth three, first eight successors per node and at most 80 nodes; divide distinct observed labels found by distinct labels in the query.
15	Local target density	Over the first ten successors, sum per-edge counts of compatible target-window labels; divide by observation positions and cap at one.
16	Current-edge evidence	Half the largest evidence between current edge and a distinct local-window label; zero initially or for an empty window.
17–18	Bounded search	Eight-simulation, depth-six value; and one minus the capped standard deviation of its latest eight values.
19	Stagnation	Consecutive nonprogress steps divided by 50.
20	Unique visits	Distinct visited nodes divided by visited nodes.

TABLE S3
TRAINING REWARD, MASKS, AND BOUNDARIES.

Item	Exact rule
SVAA/goal	Add 2.2 per newly matched observation interval. Add the decrease in straight-line goal distance divided by 120 m and clipped to $[-0.6, 0.8]$.
Lookahead/evidence	Average successor-state components 13–15. Add the bounded-search value from Section E, already in $[-1, 1]$.
Path regularization	Start at -0.01. For path-length ratio ρ, add 0.4 when $\rho \leq 1.5$; add $0.4 - 0.8(\rho - 1.5)$ when $1.5 < \rho \leq 2.5$; otherwise add $-0.8 - \min\{4.2, 1.5(\rho - 2.5)\}$. Also subtract $\min\{4, 1.1n_{\mathrm{repeat}}\}$, where n_{repeat} counts earlier occurrences of the current node among the last 12 nodes, and subtract 1.5 for immediate reverse traversal.
Training oracle	Add 4 for a forward oracle edge, 1.2 for its reverse directed mate, and 1 for an oracle node. Edge and node bonuses are additive; forward/reverse edge bonuses are exclusive.
Terminate	Goal distance below 200 m; SVAA witness coverage reaches 60%; or invalid/dead-end action. The 200-metre rule is training-only.
Truncate	Step 1500; 50 steps without a new match or 5 m best-goal improvement; or visited-node count greater than five times observation count.
Mask	Projected path-length ratio at most 5.0 and target visited fewer than three times.

real coordinate–CID trajectories could use the same interface, but all reported policies use the generated sets. For Region A, these 1,000 trajectories are also the sole development set used to select model and method hyperparameters. The 250 field-collected test trajectories remain excluded until the configuration is fixed.

Training-only oracle.: Retain valid generated GPS^* samples in input order; if more than 35 remain, select 35 including both endpoints using equally spaced NumPy round-to-even indices. Snap points to nearest valid graph nodes under the local metric in Section A, breaking an exact tie by frozen graph iteration order, and remove consecutive duplicates. Reuse a shared cached node-pair path when present. Otherwise join pairs 1–28 by directed length-weighted shortest path; from pair 29 onward, use an existing direct edge when available and the same shortest-path routine otherwise. The cache stores at most 100,000 pairs. Equal-length shortest paths inherit the pinned graph serialization and NetworkX iteration order. Abort unless every training record yields a positive-length connected oracle. Only the resulting edge/node sets enter the reward; raw coordinates are excluded from PPO state construction after oracle preparation. Oracle construction and policy training use no timestamps.

TABLE S4
FIXED PPO CONFIGURATION SHARED BY BOTH REGIONS.

Item	Value
Model	`MaskablePPO`; separate Actor and Critic branches of widths 256 and 128 with Tanh; normalized 20-vector input; 13 action slots.
Budget	Exactly 5,000,000 interactions in four environments; 1024 steps per environment and 4096 transitions per full collection; batch size 512.
Update	Discount 0.99; GAE 0.95; PPO clip 0.20; learning rate 3×10^{-4}; five epochs; entropy coefficient 0.03; value coefficient 0.5; maximum gradient norm 1.0; target KL 0.03.
Normalization	Normalize observations, not rewards; observation clip 10; stabilizer 10^{-8}.
Binding	Checkpoint, normalization mean/variance, graph, evidence cache, training manifest, and configuration manifest form one fingerprint-matched inference artifact.

Bounded lookahead.: Use horizon 6, 8 simulations, PUCT constant 1.5, and discount 0.95. Per-step utility combines SVAA witness increment, clipped goal progress, compatible labels in the local window, detour excess, and target revisit count with coefficients 1, 0.8, 0.6, -0.5, and -0.5. Add one, lower-bound at 0.05, and normalize for the prior. Each simulation follows its tree to the first unexpanded child and then greedily completes the remaining depth. Discount, divide by six, clip to $[-1, 1]$, and back up. Return the mean value of the most-visited root child; ties use the first frozen action slot. Actor, Critic, timestamps, and GPS are not called.

F. ONLINE TIME-AWARE SVAA/PPO-GUIDED MCTS

For a nonempty query sequence, define the trip duration

$$\Delta t_{\mathrm{trip}} = \begin{cases} \max\{t_n - t_1, 1\ \mathrm{s}\}, & n \geq 2,\ t_1, t_n \text{ finite}, \\ 1\ \mathrm{s}, & \text{otherwise}. \end{cases}$$

The decoder rejects an empty observation sequence. The same Δt_{trip} is used by the partial-path cap and the complete-trip filter below; timestamp ratios used by the soft corridor have their own fallback rule.

F.1 Legal Expansion, Prior, Rollout, and Backup

An action is legal only if its target can reach the goal, its projected completion satisfies the temporal length cap in F.2, its prefix remains within three times the embedded shortest-road distance, and its target has appeared fewer than three times. Use maximum depth 500 and stop after 50 steps without either a new match or 5 m best-goal improvement.

For legal action a, the unnormalized action logit is

$$\begin{aligned} z(a) = {} & 0.55A_a + 0.05E_a - 4R_a \\ & + \frac{0.25}{0.90}\big(0.10G_a + 0.18C_a \\ & \qquad + 0.20T_a + 0.24H_a\big), \\ P(a \mid s) = {} & \frac{\exp z(a)}{\sum_{b \in \mathcal{A}(s)} \exp z(b)}. \end{aligned} \tag{1}$$

Here A_a is the frozen masked-Actor probability. G_a is the decrease in shortest remaining road distance divided by the previous distance (denominator at least 1 m), clipped to $[0, 1]$. E_a is the larger cached score of the action edge for the observation at the current pointer and, when present, its immediate successor; at the last observation only that observation is used, and at the completion sentinel $E_a = 0$. C_a is the action-level corridor support defined in F.2. T_a is the new pair-time support when the action increases witness cardinality and thereby creates an adjacent matched pair; an interval extension gives $T_a = 0$. H_a is the mean successor lookahead value in state components 13–15, and R_a indicates that the target node already appears in the path history. These definitions reproduce the implemented coefficient construction without an additional prior normalization.

At a rollout endpoint, let $L_0 = \max\{L_{\mathrm{shortest}}, 1\ \mathrm{m}\}$, let d_{rem} be its shortest remaining road distance, and let L_{prefix} be its accumulated length. The previously implicit leaf terms are

$$\begin{aligned} S_{\mathrm{goal}} &= \mathrm{clip}\left(1 - \frac{d_{\mathrm{rem}}}{L_0}, 0, 1\right), \\ S^{\mathrm{leaf}}_{\mathrm{detour}} &= \mathrm{clip}\left(\frac{L_{\mathrm{prefix}}/L_0 - 1}{2}, 0, 1\right). \end{aligned} \tag{2}$$

The interval witness, edge evidence, and raw corridor utility are recomputed at that endpoint before applying the leaf formula in Table S5; V_ϕ is clipped to $[-1, 1]$.

F.2 SVAA, Temporal Constraints, and Corridor

Canonical SVAA witness.: For path $P = (e_1, \ldots, e_m)$ and nonempty observation sequence $C = (c_1, \ldots, c_n)$, initialize $D_{0,i} = D_{q,0} = 0$ and compute

$$\begin{aligned} D_{q,i} = \max\big\{ & D_{q-1,i},\ D_{q,i-1}, \\ & D_{q-1,i-1} + \mathbf{1}[c_i \in \Phi(e_q)]\big\}. \end{aligned} \tag{3}$$

Thus $K = D_{m,n}$. Backtracking follows only predecessors attaining the cell optimum and uses the fixed priority compatible diagonal, edge skip, then observation skip. Reversing the backtracked list yields ordered anchors $((i_r, a_r))_{r=1}^{K}$.

For anchor a_r, the permitted expansion bounds are the first edge when $r = 1$, otherwise $a_{r-1} + 1$, and the final edge when $r = K$, otherwise $a_{r+1} - 1$. Within those bounds, take the maximal contiguous compatible run containing a_r. Process anchors left-to-right; if raw runs overlap, move the later run's start to one edge after the previously assigned end. The earlier observation therefore owns the overlap, every anchor remains inside a nonempty interval, and two observations never reuse an edge position. Unmatched observations and connecting edges remain allowed. Expanding an interval changes its boundaries but never K. This same canonical extractor is used by state construction, search, and ranking; it does not choose a different same-cardinality witness to improve evidence or time for an individual candidate. Interval evidence is zero for $K = 0$; otherwise it averages edges within each interval and then intervals equally. Sequence coverage is $S_{\mathrm{seq}} = K/n$.

Offline training flow. Validate graph actions and evidence; construct the training-only generated-coordinate oracle; run four parallel environments; after every accepted edge recompute the SVAA witness, raw state, and bounded-search value; apply masking, reward, and episode boundaries; normalize observations; update MaskablePPO to the exact interaction budget; save the model, normalizer, graph, evidence, and manifests as one bound artifact.

Fig. S1. Complete PPO policy-training procedure.

TABLE S5
ONLINE SEARCH CONFIGURATION.

Item	Deterministic rule
Action prior	Compute the exact legal-action logit in Eq. (1) and softmax only over legal actions. The Critic has coefficient zero in this prior and is used only in leaf evaluation.
Root/PUCT	Mix 92% model prior with 8% Dirichlet noise of concentration 0.25 when multiple root actions exist. Elsewhere use no noise. PUCT constant is 1.55; exact ties use the smallest canonical edge identifier.
Expansion	Instantiate every legal child (at most 13), ordered by decreasing prior then canonical ID. Sample uniformly with probability 0.08; otherwise raise priors to 0.625, renormalize, and sample.
Rollout/backup	Greedily roll out at most 6 off-tree steps by online prior. Evaluate the rollout endpoint and back up one undiscounted scalar running mean. Rollout edges never enter the output tree.
Leaf value	$18S_{\text{align}} + 14S_{\text{edge}} + 8.5S^{\text{raw}}_{\text{corridor}} + 2S_{\text{goal}} - 6S_{\text{detour}} + 0.15\ \text{clip}(V_\phi, -1, 1)$.
Budget	30,000 simulations per query.

Temporal scopes.: For adjacent intervals $(i_r, [\ell_r, u_r])$ and $(i_{r+1}, [\ell_{r+1}, u_{r+1}])$, include both boundary edges and define the span speed in kilometres per hour as

$$\widehat{v}_r = 3.6\,\frac{\sum_{h=u_r}^{\ell_{r+1}} l(e_h)}{\max\{t_{i_{r+1}} - t_{i_r}, 1\ \text{s}\}}.$$

Soft pair compatibility. When either timestamp is nonfinite or the gap is nonpositive, set $\kappa_r = 0$. Otherwise,

$$\kappa_r = \begin{cases} \exp[-(\widehat{v}_r - 50)^2/(2\cdot 10^2)], & 40 \le \widehat{v}_r \le 60, \\ 0, & \text{otherwise.} \end{cases}$$

$$\bar{S}_{\text{pair}} = \begin{cases} 0, & K < 2, \\ \frac{1}{K-1}\sum_{r=1}^{K-1} \kappa_r, & K \ge 2. \end{cases}$$

The exact alignment score is

$$S_{\text{align}} = 0.70\, S_{\text{seq}} + 0.30\, \bar{S}_{\text{pair}}.$$

This interval-span kernel is a soft protocol proxy, not an instantaneous speed estimate or a hard feasibility rule. Thus the 40–60 km/h gate changes only the pair score and never prunes a candidate.

Partial-path hard pruning. Prefix length plus shortest remaining road distance not to exceed Δt_{trip} times 22.222222 m/s (80 km/h). *Complete-trip hard feasibility.* Complete candidates must lie within Δt_{trip} times 0.975 to 24.3 m/s, the configured 1.5–18.0 m/s range with 35% tolerance. The stricter generator cap therefore determines the effective upper bound.

CID–time corridor.: When all timestamps are finite and $n \ge 2$, use $\rho_i = \text{clip}((t_i - t_1)/\max\{t_n - t_1, 1\ \text{s}\}, 0, 1)$. Otherwise use $\rho_i = (i-1)/\max\{n-1, 1\}$ for the whole stream. Form windows starting at positions $1, 4, 7, \ldots$, each with at most ten observations, and discard a final singleton. For window w, aggregate

$$g_w(e) = \sum_{i \in w} E_{\text{CID}}(e, c_i),$$

counting repeated observation positions separately. An observation is evidence-bearing when its CID has at least one cached edge. Sort edges by decreasing $g_w(e)$ and then canonical edge identifier; retain at most 18 whose aggregate is at least half the window maximum. With b_w the number of evidence-bearing positions, set

$$r_w = \min\left\{1, \frac{\max_e g_w(e)}{\max\{1, b_w\}}\right\}$$

and retain the window only when $r_w \ge 0.24$.

For a positive-length candidate path, let $L(P) = \sum_{q=1}^{m} l(e_q)$ and parameterize edge ends by normalized cumulative length $\bar{L}_q = \sum_{h=1}^{q} l(e_h)/L(P)$. For w, draw $\text{clip}(|w|, 5, 16)$ inclusive, uniformly spaced ratios between the window's first and last ρ_i. Map each target ratio ρ to the first edge e_q with $\bar{L}_q \ge \rho$, equivalently whose cumulative length reaches $\rho L(P)$, so an exact boundary belongs to the earlier edge. For an empty or zero-length path, set all path-level corridor support terms to zero. Let h_w be the fraction of sampled edges in retained set E_w, and define

$$i_w = \min\left\{1, \frac{|E(P) \cap E_w|}{\max\{1, \min(3, |E_w|)\}}\right\}.$$

With means over retained windows, define

$$\begin{aligned} A &= \text{mean}(r_w h_w), \\ B &= \text{mean}\left(r_w \frac{(0.34 - h_w)_+}{0.34}\right), \\ C &= \frac{\sum_w r_w i_w}{\sum_w r_w}, \\ \bar{h} &= \text{mean}(h_w), \\ b(\bar{h}) &= \begin{cases} -5.20(0.30 - \bar{h})/0.30, & \bar{h} < 0.30, \\ 0.80(\bar{h} - 0.30), & \bar{h} \ge 0.30, \end{cases} \\ S^{\text{raw}}_{\text{corridor}} &= 0.75A - 3.50B + 2.60C + b(\bar{h}). \end{aligned}$$

All values are zero if no window survives. For action a at the current observation pointer k, let $\mathcal{W}(k)$ be the retained windows containing k and define the action-level term used in Eq. (1) by

$$C_a = \begin{cases} \dfrac{\sum_{w\in\mathcal{W}(k)} r_w \mathbf{1}[a \in E_w]}{\sum_{w\in\mathcal{W}(k)} r_w}, & \sum_{w\in\mathcal{W}(k)} r_w > 0, \\ 0, & \text{otherwise.} \end{cases}$$

After duplicate coordinates at edge joins are removed, at most 900 uniformly spaced path coordinates are inspected only to reject nonfinite geometry or a coordinate without its canonical-edge association; this check contributes no score, and i_w always uses the complete canonical edge set. The corridor affects priors and scores but never creates, completes, repairs, or hard-prunes a path.

F.3 Complete Candidates, Normalization, and Final Ranking

Traverse MCTS children in canonical depth-first order after 30,000 simulations. Keep only complete tree-generated goal paths satisfying the trip filter and deduplicate by full canonical edge sequence. Do not complete prefixes with shortest paths, corridor edges, or rollout edges. If none remain, report failure. Normalize only raw corridor utility within the query's retained candidate set: use min–max normalization when its range exceeds 10^{-12}, and otherwise clip each raw value to $[0, 1]$. With $L_0 = \max\{L_{\text{shortest}}, 1\ \text{m}\}$, define $S_{\text{detour}} = \text{clip}((L/L_0 - 1)/2, 0, 1)$,

$$\begin{aligned}\text{Score}_{\text{final}} = S_{\text{align}} &+ 0.75 S_{\text{edge}} \\ &+ 0.75 S^{\text{norm}}_{\text{corridor}} - 0.45 S_{\text{detour}}.\end{aligned}$$

The final score contains neither Critic, an extra speed term, nor an evaluation metric. Exact ties choose the lexicographically smallest complete canonical edge sequence.

G. Controlled Comparisons and Compact Results

G.1 Baseline and Ablation Definitions

Baselines.: O–D Dijkstra uses endpoints only. Yen K-Shortest Paths generates at most 200 loopless paths; Beam Search retains width 128 and uses a 30,000-state expansion cap. Both use the common road–CID evidence, SVAA witness, temporal constraints, and final route score specified below, without learned Actor priors or Critic values. CID-ST-HMM uses support emissions and timestamp-, reachability-, and endpoint-aware Viterbi. CID-MMA Transformer predicts candidate-road probabilities followed by graph-constrained Viterbi. The three ODT adaptations in Table S6 retain selected architectural components but change information sources, supervision, or decoding. They are not native-system reproductions.

Adaptation scope.: The following implementations retain selected architectural components but replace native information sources or supervision to fit the ODT task. They are not reproductions of the original published systems.

Common route-evaluation contract.: Yen, Beam, and the proposed decoder share the frozen E_{CID}, strict compatibility threshold 0.18, canonical SVAA witness in F.2, pair-time kernel, CID–time corridor, and detour definition. Let $\mathcal{C}_A(X)$ be the retained feasible complete-candidate set of method $A \in \{\text{Yen}, \text{Beam}, \text{MCTS}\}$. All three methods rank their complete candidates using the F.3 score:

$$\begin{aligned}\text{Score}_{\text{final}}(P; \mathcal{C}_A) = 0.70 S_{\text{seq}} + 0.30 \bar{S}_{\text{pair}} + 0.75 S_{\text{edge}} \\ + 0.75 S^{\text{norm}}_{\text{corridor}}(P; \mathcal{C}_A) - 0.45 S_{\text{detour}}.\end{aligned}$$

Raw corridor utility is normalized within each method's own retained complete-candidate set using the F.3 rule. Thus the evidence, alignment, temporal terms, and final coefficients are shared, while each method constructs its own candidates. Yen and Beam use no learned Actor priors or Critic values.

Method-specific budget semantics.: The 30,000 MCTS simulations count selection, expansion/rollout, and backup iterations, not 30,000 complete routes. Yen's $K = 200$ caps generated complete O–D paths, whereas Beam width 128 caps retained partial paths at each depth. These controls are neither candidate-count nor compute equivalents. We therefore compare end-to-end recovery accuracy under pre-frozen configurations, not matched-runtime efficiency. The task is retrospective reconstruction: "online decoding" means query-time use of frozen artifacts, not a real-time streaming requirement, and latency is not a claimed outcome.

Controlled interventions and ablations.: The first four variants retain the complete method's frozen PPO checkpoint and normalizer and change one decoder-side rule; they measure frozen-guidance sensitivity, not the effect of retraining an alternative policy. *Synchronous Alignment* replaces only SVAA with position-wise matching: road and observation positions advance together, without gaps or multi-edge intervals. *One-CID-One-Edge* retains, for each CID, only its highest-evidence compatible road edge (canonical-edge tie-break). *Point/Centroid Representation* replaces each empirical support Ω_c by the mean accepted coordinate and rebuilds compatibility. *No Set-Valued Alignment* instead retains, for each edge, only its highest-evidence compatible CID (CID-string tie-break), while a CID may still label many edges. Thus the latter two relation collapses are not duplicates. *Pure PPO* removes MCTS, rollouts, and complete-tree comparison and greedily selects the highest-probability legal Actor action. *Pure MCTS* retains the same tree, budget, hard rules, extraction, and ranking but removes Actor priors and Critic leaf values. *No Timestamp* removes every timestamp-derived online speed, trip, corridor-placement, leaf, and ranking term while retaining CID order. All unlisted graph, split, endpoint, metric, budget, seed, and tie settings remain fixed.

G.2 Region A Protocol and Results

Region A has 92,232 field-collected calibration records covering 911 normalized raw-CID values and a graph with 6,277 nodes and 12,777 directed edges. Its 1,000 policy-training trajectories are generated from those records and the graph under Section E, requiring no additional training

F.4 End-to-End Textual Pseudocode

Input: directed graph and endpoint-embedding rules; origin and destination; nonempty timestamped raw-CID sequence; frozen evidence cache, PPO Actor/Critic, normalizer, and inference configuration. **Output:** one canonical connected O–D path or explicit failure.

1) Validate all artifact fingerprints and the 13-slot action contract. Embed the start and then the goal, compute L_0 and Δt_{trip}, and build the deterministic CID–time corridor.
2) Initialize the root with an empty edge sequence, zero visits, and its canonical SVAA witness, state, legal-action mask, length, visit history, and stagnation variables; initialize every subsequently created tree edge with $N(s,a) = Q(s,a) = 0$.
3) For each of 30,000 simulations, descend through expanded nonterminal nodes by the main-paper PUCT rule. Terminate a branch at the depth or stagnation limit; otherwise mask actions that violate reachability, projected-completion length, detour, or revisit rules.
4) At the first expandable node, construct the normalized PPO state, evaluate Eq. (1), instantiate every legal child in prior order, mix Dirichlet noise only at the root, and sample one child by the fixed uniform/power-mixture rule.
5) Append the sampled tree edge; update length and visit history; recompute the complete canonical SVAA witness, distinguishing extension of an existing interval from a new matched observation; then update pair-time and stagnation state.
6) Unless terminal, run at most six off-tree greedy steps using the same legal mask and online prior. Rollout edges update the hypothetical state but are never inserted into the output tree.
7) At the rollout endpoint, recompute S_{align}, S_{edge}, $S^{\mathrm{raw}}_{\mathrm{corridor}}$, S_{goal}, $S^{\mathrm{leaf}}_{\mathrm{detour}}$, and the clipped Critic term. Apply the leaf formula in Table S5 and update the running mean V at every visited non-root tree edge by $N \leftarrow N + 1$ and $Q \leftarrow Q + (V - Q)/N$.
8) After the budget, traverse tree children in canonical depth-first order. Retain only tree-generated goal paths satisfying the complete-trip filter and deduplicate by their full canonical edge sequence; report failure if the set is empty.
9) Recompute every retained path's canonical witness and final components. Normalize only $S^{\mathrm{raw}}_{\mathrm{corridor}}$ within this candidate set and apply the F.3 score.
10) Return the highest-scoring path; an exact score tie selects the lexicographically smallest complete canonical edge sequence. No prefix completion, rollout edge, corridor edge, or reference path can enter the returned sequence, and the final ranking contains no Critic term.

Fig. S2. Complete online constrained-decoding procedure.

TABLE S6
ARCHITECTURAL SCOPE OF THE IMPLEMENTED ODT ADAPTATIONS.

Adaptation	Retained components	Adapted information or decoding
DMM-ODT	CID/road embeddings, GRU encoder–decoder, global attention	Regional road supervision, O–D conditioning, timestamp-aware decoding, and graph stitching replace parts of the native pipeline.
CID-LHMM-ODT	Multi-relational representations, learned probabilities, Viterbi decoding	Calibration-derived observations replace tower-position information; O–D/time constraints and graph stitching are added.
ZSMM-ODT-Sup	Two-channel raster representation, U-Net calibration, constrained path solving	Calibration-derived CID positions and regional road supervision replace the native zero-shot evaluation setting.

drives; they are the only trajectories used for policy fitting and hyperparameter selection. All 250 test trajectories are real vehicle drives recorded with the probe and held out until the model and hyperparameters are fixed. They are used only for final held-out evaluation and never form a model-selection split. A different researcher collected them without access to the sampled training O–D pairs, generated routes, CID sequences, or other training-set information. Test records do not enter calibration, training-data generation, policy training, or hyperparameter selection. Test GPS, CID, and timestamps were recorded synchronously. Evaluation GPS polylines are converted independently of predictions into canonical road-edge references by approximately 5-metre densification, nearest-node mapping, and directed length-weighted shortest paths between nonadjacent mapped nodes. These automatically constructed references are used only after prediction and may retain projection uncertainty at parallel or grade-separated roads.

Against Beam Search, paired 10,000-resample 95% intervals are $[18.15, 23.29]$ for road-edge F1 and $[20.23, 25.94]$ for length-weighted F1; Ours wins 232/250, ties one, and loses 17. These intervals reflect test-query sampling uncertainty.

TABLE S7

MEAN REGION A RESULTS OVER ALL 250 HELD-OUT TRAJECTORIES (%). F_{LCS} IS THE ORDER-SENSITIVE ROAD-EDGE SEQUENCE LCS-F1.

Method	P	R	F	P^L	R^L	F^L	S_{OD}	F_{LCS}
Ours	**90.25**	**95.30**	**92.47**	**92.08**	**96.03**	**93.83**	**100.00**	**88.95**
O–D Dijkstra	84.94	61.97	69.85	84.87	61.18	68.81	99.60	75.37
Yen KSP + score	75.68	61.82	66.31	76.39	60.77	65.60	99.60	67.27
Beam Search + score	80.31	67.85	71.79	80.11	66.89	70.79	99.60	74.34
CID-ST-HMM	27.23	19.71	18.92	23.92	20.59	18.24	46.40	29.56
CID-MMA Transformer	26.26	20.29	18.65	24.68	21.00	18.70	45.60	30.18
DMM-ODT	55.00	32.40	37.32	53.38	30.51	34.80	100.00	56.20
CID-LHMM-ODT	63.87	41.33	45.54	60.00	42.96	45.57	100.00	67.41
ZSMM-ODT-Sup	70.17	61.21	61.74	74.54	55.77	59.93	100.00	51.84
Synchronous Alignment	74.44	71.59	71.52	74.90	70.90	70.82	100.00	69.40
One-CID-One-Edge	77.27	70.98	72.19	77.75	71.04	72.10	100.00	69.34
Point/Centroid	81.23	76.37	76.72	82.12	76.10	76.68	100.00	75.15
No Set-Valued Alignment	77.24	69.86	71.65	78.26	70.05	71.93	100.00	71.11
Pure PPO	31.93	64.60	38.35	31.42	63.39	37.26	87.20	15.76
Pure MCTS	76.72	77.32	75.61	77.43	77.13	75.59	89.20	15.93
No Timestamp	78.59	56.74	58.88	77.66	57.07	58.66	90.40	15.42
No-GPS Training	82.38	94.26	86.65	83.58	94.68	87.31	100.00	81.29

For a predicted edge sequence E_p and reference sequence E_r, $F_{\mathrm{LCS}} = 2\,\mathrm{LCS}(E_p, E_r)/(|E_p| + |E_r|)$; unlike edge-set F, it preserves traversal order and repeated nonconsecutive edges. The additional *No-GPS Training* control removes the GPS^*-derived oracle reward during policy training while retaining the calibrated road–CID evidence and endpoint inputs. A separate PPO checkpoint and observation normalizer are fitted for this training setting and frozen before evaluation. The online query inputs, constrained decoder, and evaluation protocol remain unchanged.

H. FIGURES REFERENCED BY THE MAIN PAPER

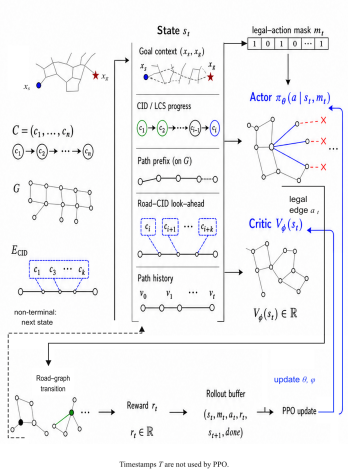


Fig. S3. PPO neighbor-edge policy learning. Timestamps are excluded from state, action, and reward; the Actor, Critic, and normalizer are frozen before online decoding.

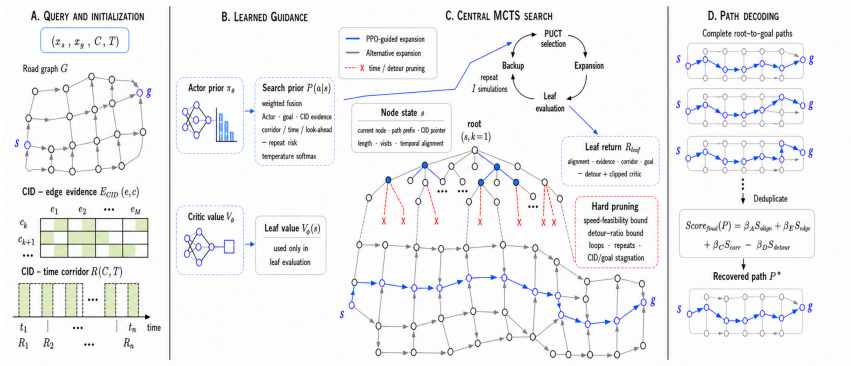


Fig. S4. Online time-aware PPO-guided MCTS. Only complete tree paths are candidates; rollout edges estimate leaf value but never enter an output path, and the Critic does not enter final ranking.

H.1 Fixed Qualitative and Per-Trajectory Audits

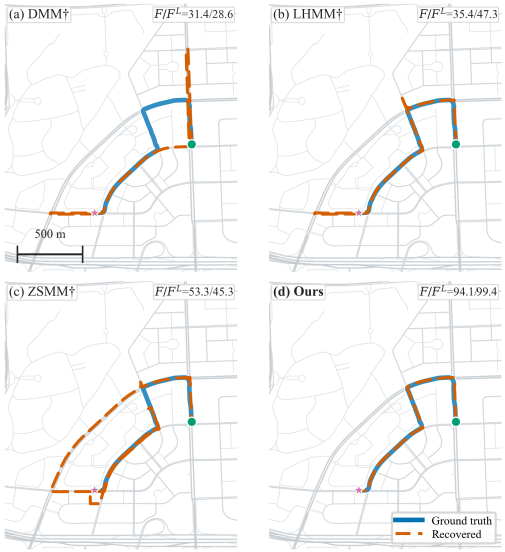


Fig. S5. Same-query comparison on fixed Region A test trajectory 9: DMM-ODT, CID-LHMM-ODT, ZSMM-ODT-Sup, and Ours. Every panel uses the identical query and automatically constructed ground truth; blue is ground truth and red is the method's single returned path. No method is rerun or manually selected for this trajectory.

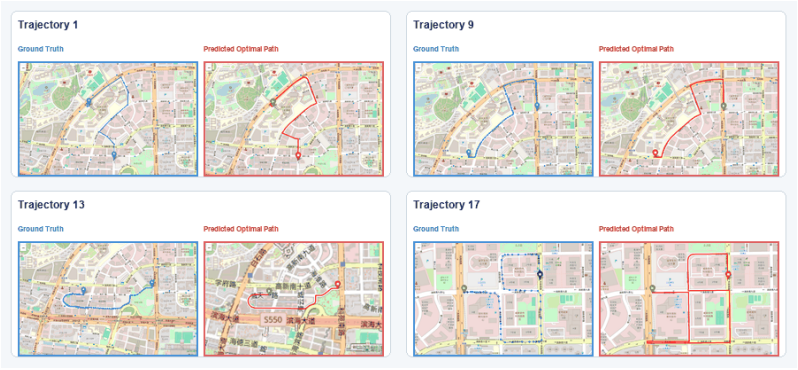


Fig. S6. Reference and recovered routes for fixed Region A examples (trajectories 1, 9, 13, and 17). The prediction is the single returned path from the fixed run; no trajectory-specific rerunning or manual path selection is used.



Fig. S7. Compact per-trajectory audit for Ours and all seven controlled variants on the 250 Region A tests in fixed original order. Columns are identical trajectories across rows; color encodes stored road-edge F1 and a white cross marks O–D failure. The visualization uses all 2,000 per-trajectory records without sorting or selection; the seven-metric macro averages are in Table S7.

Reading guide. The training-only oracle is separate from the PPO state and saved inference artifact. Tree edges can form candidates; rollout edges only estimate leaf value; corridor edges only provide soft evidence. "Ground Truth" means the automatically constructed reference; "Predicted Optimal Path" means the single returned candidate from the frozen run. Background maps never enter decoding or metrics.

Controlled interpretation. The four representation/SVAA interventions retain 100% connectivity but lose 15.75–20.95 road-edge-F1 points, separating route fidelity from path feasibility. Pure PPO exposes long-horizon local-policy error; Pure MCTS shows the benefit of learned guidance; and No Timestamp shows that temporal constraints affect recall and feasibility. These controlled changes do not rank native systems under their original supervision.

Connectivity results. O–D Dijkstra, Yen KSP, and Beam Search each return a valid canonical O–D path on 249 of the 250 Region A queries, yielding $S_{\mathrm{OD}} = 99.60\%$. The proposed decoder returns valid connected paths on all 250 queries. All queries, including failed recoveries, contribute to the reported macro-averaged metrics.

I. Separate Region B Evaluation

Region B uses a separately prepared regional pipeline with 1,000 training trajectories generated from its own calibration corpus and road graph under Section E, 1,316,400 calibration records, 3,291 unique raw-CID values, a graph of 2,152 nodes and 4,225 directed edges, and 250 real vehicle-collected test trajectories. A different researcher collected the tests without access to the Region B training-set information, and test records do not enter calibration, training-data generation, policy training, or hyperparameter selection. Its regional policy is fitted on the 1,000 generated trajectories using the Region A hyperparameters and decoding settings; those settings are fixed before the 250 real tests are evaluated. Region B otherwise uses distinct regional artifacts. It replicates the complete calibration-and-training pipeline rather than performing zero-shot transfer.

TABLE S8
Scale of the Region B data and graph.

Asset	Count
Generated training trajectories	1,000
Calibration records	1,316,400
Unique raw-CID values	3,291
Road-graph nodes	2,152
Directed road edges	4,225
Vehicle-collected test trajectories	250

TABLE S9
Aggregate results for the two separate regional evaluations (%).

Region	P	R	F	S_{OD}
A	90.25	95.30	92.47	100.00
B	96.19	93.94	94.86	100.00

Region	P^L	R^L	F^L
A	92.08	96.03	93.83
B	96.67	94.43	95.34

Evaluation and interpretation.: Region B uses the same per-trajectory macro metrics, failure conventions, canonical-edge restoration, and evaluation-reference construction as Region A. A failed query remains in every denominator. The Region B row does not enter the Region A baseline gains or bootstrap intervals. Region B combines independently constructed regional evidence and a separately trained policy with the Region A hyperparameters and decoding settings. Its results demonstrate replication of the complete calibration–training–decoding pipeline across two separately prepared regional evaluations under a common configuration.

Data efficiency and fixed-seed repeatability.: For each calibration-data fraction in Figure S8(a), we construct the regional road–CID evidence from the corresponding subset and retrain the policy under the same configuration. As the fraction increases from 10% to 100%, road-edge F1 improves from 90.16% to 94.86%, and length-weighted F1 improves from 90.60% to 95.34%. These results demonstrate improved end-to-end recovery performance with larger regional calibration corpora. Panel (b) compares policies trained with 50, 250, 500, and 1,000 trajectories. After training, all four settings are evaluated on exactly the same 250 test trajectories using the same road graph, road–CID evidence, constrained decoder, and search budget. Thus, the observed differences reflect the training data and the resulting policy, rather than differences in the evaluation set. The 50- and 250-trajectory settings already learn useful local preferences over neighboring road edges, while topology, endpoint constraints, road–CID evidence, and temporal feasibility continue to strongly constrain online decoding. A relatively small training set can therefore still provide effective guidance to the complete decoder. The temporary reduction at 500 trajectories indicates that performance need not vary monotonically with training-set size. Under the fixed optimization budget and unchanged training configuration, the additional trajectories may introduce more ambiguous Cell-ID–road associations and a broader state–action distribution while reducing the effective optimization exposure available to each recurring pattern. The intermediate-scale policy may consequently be less fully optimized. With the complete 1,000-trajectory set, broader coverage of road structures and Cell-ID combinations provides sufficient diversity for the generalization benefit to dominate, yielding the best F and F^L. Because each size corresponds to one trained policy, the local reduction at 500 trajectories is interpreted as training-composition and optimization sensitivity under the fixed budget rather than as a uniquely identified causal effect or evidence that additional training data are intrinsically harmful. Panel (c) evaluates fixed-seed repeatability on Region A. Before prediction, 20 trajectories are randomly sampled without replacement from the 250-test set, independently of prediction outcomes. Each is predicted ten times using the same frozen model, input observations, inference configuration, evaluation procedure, and inference random seed, yielding 200 predictions. All reported metrics remain identical across repetitions, with zero within-trajectory sample standard deviations. This demonstrates repeatable execution of the frozen inference pipeline under the fixed-seed protocol. Main effectiveness results and method comparisons continue to use all 250 test trajectories.

Qualitative audit.: The complete fixed-order visual audit contains 13 consecutive sheets: 12 sheets of 20 test trajectories and one sheet for trajectories 241–250. Every card uses the same extent for reference and prediction. All 13 sheets, in original test order, are packaged in the separate anonymous supplementary media archive; Figure S9 shows the first and final sheets, selected by position in that fixed order rather than performance.

Regional invariance audit.: The two regional runs share the evidence-grid and buffering constants, direction-reliability rule, sparsification cap and compatibility threshold, graph canonicalization and 13-slot action contract, state and reward definitions, PPO architecture and optimizer settings, training interaction budget, online simulation budget, PUCT and rollout rules, temporal bounds, corridor construction, final coefficients, and deterministic tie rules. No Region B threshold, coefficient, budget, or speed range was selected from its test results. Region-specific quantities are limited to the permitted inputs and

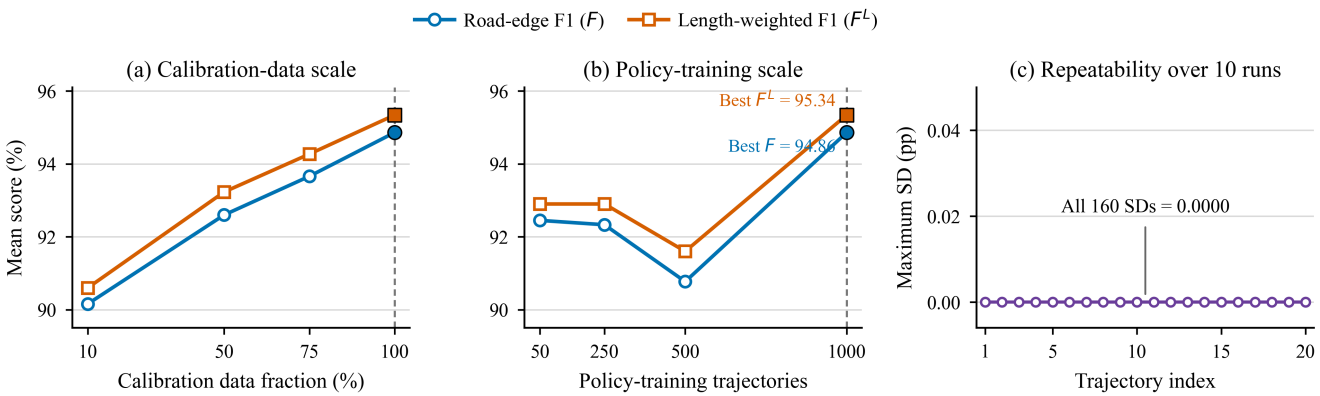


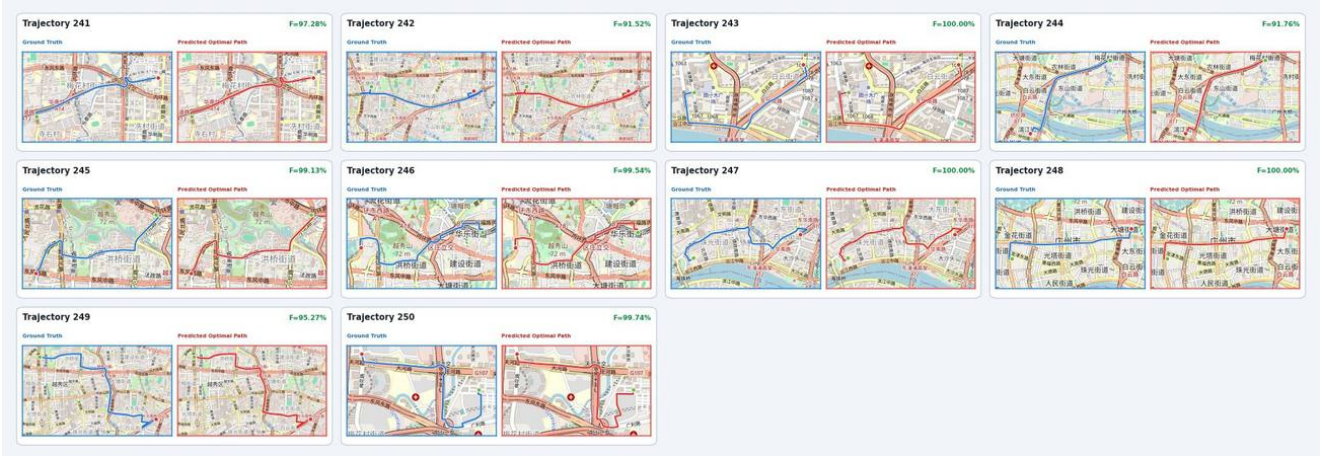
Fig. S8. Data efficiency and fixed-seed repeatability. (a) Region B calibration-data scale, with regional evidence construction and policy retraining for each subset; (b) Region B policy-training-data scale; and (c) sample standard deviations over ten predictions using the same inference random seed for each of 20 trajectories randomly sampled without replacement from the 250-test Region A set.

their consequences: calibration records, road graph, generated-training and held-out-test sets, cache/hash values, and the PPO weights and normalization statistics learned from those inputs.

Artifact and reference separation.: The Region B evidence cache is built before evaluation from its own permitted calibration corpus and is consumed read-only by training and prediction. Its field-collected test GPS polyline is projected to a reference only for post-prediction metric computation. Neither that polyline nor the resulting canonical edges can change search, corridor construction, candidate extraction, or ranking. This is the same separation enforced in Region A and prevents the additional regional experiment from silently changing the method's information boundary.



(a) Trajectories 1–20



(b) Trajectories 241–250

Fig. S9. First and final fixed-order Region B route sheets. Blue denotes the automatically constructed reference and red the single path from the frozen inference run. The other 11 consecutive sheets are in the anonymous media archive; no query is rerun and no path is manually selected. Pixel overlap is not an evaluation metric; Table S9 is computed from canonical edges and lengths.

J. Reproducibility Checklist and Limitations

J.1 Reviewer-Facing Reproducibility Checklist

TABLE S10
REVIEWER-FACING PHASE CHECKLIST.

Phase	Fixed boundary
Calibration	Validate the graph, accepted probe GPS–CID rows, generator constants, sorted reductions, and hashes. Exclude tower coordinates/inventory, operator road associations, and evaluation queries.
PPO training	Use the frozen graph/cache, calibration-derived, road-graph-generated training trajectories, training-only oracle, 20-vector state, mask, and exact budget for policy fitting and hyperparameter selection. Exclude all 250 held-out real tests and their timestamps from model selection, state, action, reward, and bounded lookahead.
Online decoding	Use endpoints, timestamped CIDs, graph/cache, checkpoint, normalizer, and fixed search configuration. Exclude intermediate query GPS, evaluation edges, post-hoc completion, and manual repair.
Evaluation	Compare the returned canonical path with an independently constructed held-out reference. Compute per-query metrics before macro-averaging and provide no feedback to any earlier phase. This empirical validation is not a model-selection split.

Every proposed result is bound to a regional evidence fingerprint, graph, PPO checkpoint, matching normalizer, split manifest, and inference configuration. The four representation/alignment interventions reuse the complete method's Region A PPO checkpoint and normalizer and change only the named decoder-side rule. The additional training-supervision control uses its separately fitted checkpoint and normalizer. Displayed percentages are rounded only after aggregation. The separate media archive contains only fixed rendered outputs and a manifest; it does not alter the technical PDF or provide a mutable web pointer.

J.2 Limitations and Future Work

The evaluated pipeline uses regional GPS–CID calibration, policy training, known endpoints, and temporal settings fixed for the vehicle acquisitions. Future acquisitions can test perturbed endpoints, different traffic and sampling regimes, evidence aging, and reduced-calibration transfer. Metrics use automatically constructed road-edge references; targeted audits can quantify ambiguity near parallel or grade-separated roads. Query-bootstrap intervals and fixed-seed repetitions characterize distinct sources of uncertainty, while independent training and inference seeds can extend this assessment.

Finally, planned release remains subject to applicable map/data licenses and privacy review. The benchmark is generated from researcher-controlled probe drives rather than passive subscriber exports; nevertheless, timestamps and mobility traces are sensitive and should be minimized, access-controlled, or de-identified under the eventual release protocol.